

Cardiovascular disease EDA report

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Abstract

Cardiovascular disease (CVD) is one of the leading causes of death worldwide. Early identification of risk factors can help prevent severe complications. The purpose of this project is to perform an Exploratory Data Analysis (EDA) on a cardiovascular disease dataset to understand the characteristics of the data, identify relationships between variables, detect abnormal observations, and investigate factors associated with cardiovascular disease.

The analysis includes data cleaning, preprocessing, visualization, statistical hypothesis testing, and feature engineering using Python libraries such as Pandas, NumPy, Matplotlib, Seaborn, and SciPy.

The results showed that age, blood pressure, cholesterol, glucose level, and BMI were significantly associated with cardiovascular disease, while smoking showed a weaker association in this dataset. The cleaned dataset provides a reliable basis for future machine learning prediction models.

1. Introduction

Exploratory Data Analysis (EDA) is an essential step in every data science project. It allows researchers to understand the structure of the dataset before building predictive models.

The cardiovascular disease dataset contains medical information collected from patients, including demographic characteristics, physical measurements, laboratory results, lifestyle habits, and the presence or absence of cardiovascular disease.

The main objectives of this project are:

- Understand the dataset
- Clean noisy and duplicated records
- Detect abnormal values
- Explore relationships among variables
- Perform statistical hypothesis testing
- Create new meaningful features

2. Dataset Description

The dataset contains medical records of approximately **70,000 patients**.

Main variables include:

Table1. Dataset feature	
Variable	Description
Age	Patient age (days converted to years)
Gender	Male/Female
Height	Height (cm)
Weight	Weight (kg)
Ap_hi	Systolic blood pressure
Ap_lo	Diastolic blood pressure
Cholesterol	Cholesterol level
Glucose	Glucose level
Smoke	Smoking status
Alcohol	Alcohol consumption
Active	Physical activity
Cardio	Cardiovascular disease (Target variable)

Table2. Dataset information	
Property	Value
Samples	70,000
Features	12
Target	Cardio (0/1)

3. Data Preprocessing

Several preprocessing steps were applied before analysis.

3.1 Removing unnecessary columns

The **ID** column does not contain useful information for analysis. Therefore it was removed.

3.2 Missing values

The dataset was checked for missing values. No missing values were found.

3.3 Duplicate records

Duplicated rows were identified and removed to improve data quality. Overall, 24 duplicated records were removed.

3.4 Age conversion

Age was originally stored in days. It was converted into years using:

$\text{Age} = \text{Age} / 365$

This makes interpretation much easier.

3.5 Data type optimization

Column data types were optimized to reduce memory usage.

Examples:

- uint8
- int16
- float32

4. Outlier Detection

Outliers were investigated using the **Interquartile Range (IQR)** method.

Scatter plots were used together with upper and lower IQR boundaries.

However, outliers in age, height, and weight were retained because they may represent genuine physiological variations rather than measurement errors.

5. Medical Data Cleaning

Instead of relying solely on statistical methods, medical knowledge was used to remove impossible blood pressure values.

Examples removed include:

- Systolic BP < 70
- Systolic BP > 250
- Diastolic BP < 40
- Diastolic BP > 150
- Systolic BP $\leq$ Diastolic BP

This step significantly improved data quality.

6. Exploratory Data Analysis

Several visualization techniques were used.

Histogram

Used to inspect numerical variable distributions.

Variables:

- Age
- Height
- Weight
- Systolic BP
- Diastolic BP

Boxplots

Boxplots were generated to visualize:

- Median
- Quartiles
- Outliers

Countplots

Categorical variables were explored:

- Gender
- Cholesterol
- Glucose
- Smoking
- Alcohol
- Physical Activity
- Cardiovascular Disease

Correlation Heatmap

A correlation matrix was calculated.

The heatmap revealed moderate positive correlations between:

- Age
- Blood pressure
- Weight
- Cardiovascular disease

KDE Plots

Kernel Density Estimation plots compared distributions of numerical variables between healthy and diseased individuals.

Boxen Plots

Boxen plots illustrated distribution differences between healthy and diseased patients.

7. Statistical Analysis

7.1 Shapiro-Wilk Test

The Shapiro-Wilk test was applied to numerical variables.

Null hypothesis

Data follow a normal distribution.

Alternative hypothesis

Data do not follow a normal distribution.

Results indicated that none of the numerical variables followed a normal distribution.

Therefore, non-parametric statistical tests were selected.

7.2 Chi-Square Test

Chi-square tests examined associations between categorical variables and cardiovascular disease.

Variables tested:

- Gender
- Cholesterol
- Glucose
- Smoking
- Alcohol
- Physical activity

Most categorical variables showed statistically significant associations with cardiovascular disease.

7.3 Mann–Whitney U Test

Since numerical variables were non-normal, the Mann–Whitney U test compared healthy and diseased groups.

Variables tested:

- Age
- Height
- Weight
- Systolic BP
- Diastolic BP

Significant differences were observed for most variables.

8. BMI Feature Engineering

A new feature, Body Mass Index (BMI), was created.

$$\text{BMI} = \text{Weight} / \text{Height}^2$$

Patients were categorized as:

- Underweight
- Normal
- Overweight
- Obese

Two different visualizations were produced:

- Percentage bar chart
- Countplot

These analyses demonstrated that cardiovascular disease prevalence increases with BMI category.

9. Key Findings

Major observations include:

- Older individuals exhibit a higher prevalence of cardiovascular disease.
- Higher systolic and diastolic blood pressure are strongly associated with disease.
- Overweight and obese patients show substantially higher disease rates.
- Elevated cholesterol is one of the strongest indicators.
- Elevated glucose also increases disease prevalence.
- Smoking alone does not appear to be a strong predictor in this dataset because smokers represent a much smaller proportion of the sample.
- Physical inactivity is associated with increased disease occurrence.

10. Limitations

- The dataset is observational and cannot establish causality.
- Missing values were not present; therefore, imputation methods were not evaluated.
- The analysis focused on EDA and statistical inference rather than predictive modeling.
- Feature interactions were not extensively investigated.

11. Conclusion

This project successfully performed a complete Exploratory Data Analysis of a cardiovascular disease dataset.

The analysis included:

- Data cleaning
- Data preprocessing
- Outlier detection
- Medical data validation
- Visualization
- Statistical hypothesis testing
- BMI feature engineering

The findings indicate that age, blood pressure, cholesterol, glucose level, and BMI are strongly associated with cardiovascular disease.

This cleaned dataset can be used as a reliable foundation for future machine learning models aimed at cardiovascular disease prediction.

These findings are consistent with previous epidemiological studies reported in the literature.

References

1. World Health Organization (WHO)
2. Scikit-learn Documentation
3. Pandas Documentation
4. NumPy Documentation
5. Matplotlib Documentation
6. Seaborn Documentation
7. SciPy Documentation